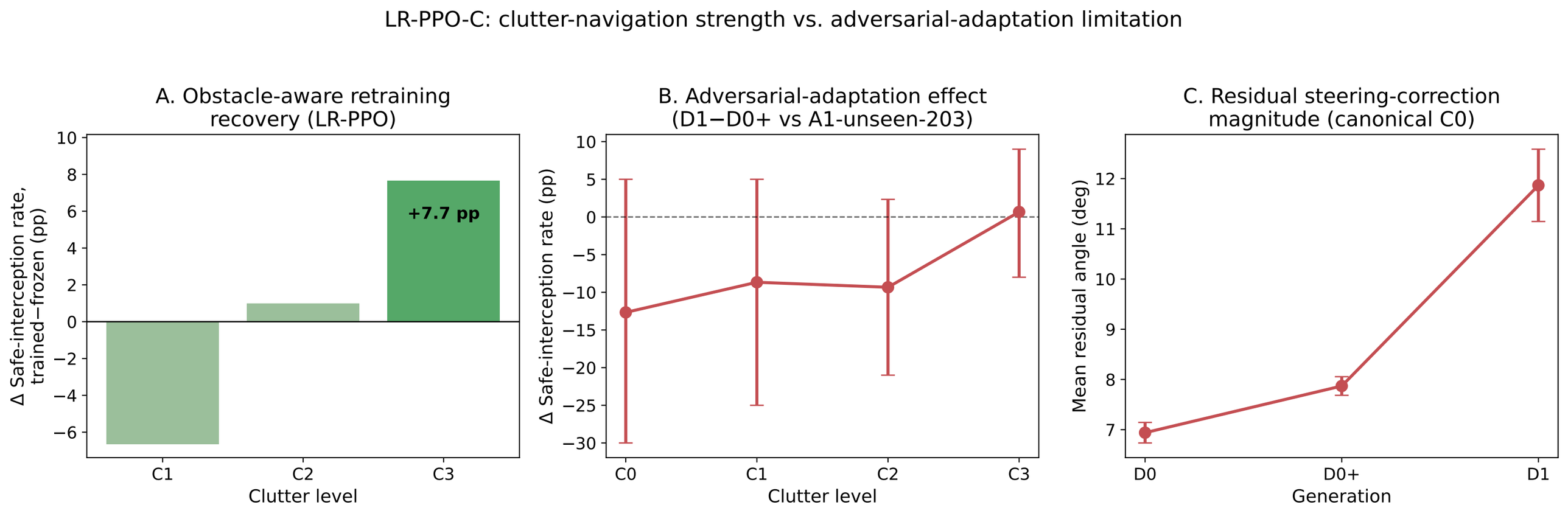
LR-PPO-C: clutter-navigation strength vs. adversarial-adaptation limitation
A. Obstacle-aware retraining recovery (LR-PPO)
Δ Safe-interception rate, trained−frozen (pp)
+7.7 pp
C1
C2
C3
Clutter level
B. Adversarial-adaptation effect (D1−D0+ vs A1-unseen-203)
Δ Safe-interception rate (pp)
C0
C1
C2
C3
Clutter level
C. Residual steering-correction magnitude (canonical C0)
Mean residual angle (deg)
D0
D0+
D1
Generation